Collin A. P. Joseph, MEng.

Artificial Intelligence Engineer | Machine Learning Engineer

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PROFESSIONAL SUMMARY

Results-driven Artificial Intelligence and Machine Learning Engineer with 4+ years of experience in Software Development and a continuous-growth mindset. Skilled in Python, Data Science, and AI Application Development. Proven track record of delivering customer-facing software products and features. Seeking to leverage experience and education to contribute to impactful AI related roles.

WORK EXPERIENCE

Technologist

Sep 2026 - Present

Seneca Polytechnic — Toronto, Ontario

- Shipping upgrades to Microsoft Foundry (Azure) based AI chat Agent using Python API, improving performance and correcting undesired behavior.
- Developing evaluation framework for AI Agent to quantify performance and consistently identify failure points.
- Upgraded production web application from Angular v19 to v22, improving application security and accessibility to new features.
- Debugged and improved web application behavior using Typescript and SQL.
- Accelerating development speed and volume using GitHub Copilot.

Software Developer

Dec 2020 - Mar 2025

Yulio Technologies Inc. — Toronto, Ontario

- Shipped 12 features across 3 key software products per year using Agile Project Management.
- Led and mentored junior developers on multi-national development team while continuing hands-on development.
- Published and maintained multi-platform Autodesk Revit application using C# (.NET Framework / .NET Core).
- Built C# REST API client to upload 3D gLTF models to cloud rendering platform.
- Deployed C++ tool to generate and manipulate HDR image-based lighting environment maps.

Software Developer
Nov 2019 - Apr 2020

wrnch AI
Montreal, Quebec

- Delivered new features for Computer Vision (pose estimation) inference server using C++ and Docker.
- Enabled product enhancement by prototyping new CNN models and updated existing ones using Python and Tensorflow.
- Automated data pre-processing pipeline and model benchmarking using Python, streamlining testing segment of development process.
- Deployed production watchdog client using Python which enhanced server reliability and control.

Graduate Student Researcher
Sep 2017 - Oct 2019

McGill University
Montreal, Quebec

- Developed ensemble SVM model using Python to classify subjects of microwave radar signals.
- Elevated predictive performance by proposing new time-frequency decomposition features.
- Improved model training speed by 60% using derivative-free optimization algorithms.
- Performed extensive statistical analysis on radar scan characteristics using MATLAB and R.
- Streamlined data pre-processing by developing tools in Python improving workflow for project team.

Signal Processing Algorithm Developer
Sep 2016 - Dec 2016

ON Semiconductor
Waterloo, Ontario

- Prototyped an environment classification algorithm for low-resource digital hearing-aids (MATLAB & Assembly).
- Optimized acoustic noise reduction algorithm in Assembly, decreasing power consumption by 20%.
- Leveraged fixed point arithmetic operations to maximize algorithm efficiency on embedded hardware.
- Executed rigorous regression testing on algorithm to ensure performance.

Signal Processing Algorithm Developer
Jan 2016 - Apr 2016

ON Semiconductor
Waterloo, Ontario

- Prototyped a digital equalizer firmware module in Assembly Language.
- Simulated and performed experimental analysis on digital equalizer using MATLAB.
- Executed and documented extensive test procedures for acoustic feedback cancellation algorithm.
- Documented and presented analysis and test results to firmware and software development team.

Signal Processing Algorithm Developer
May 2015 - Aug 2015

ON Semiconductor
Waterloo, Ontario

- Evaluated performance of static noise reduction algorithms using MATLAB.
- Developed functional simulations to evaluate performance of various DSP algorithm configurations.
- Performed experimental analysis on multiple noise estimation methods.
- Developed and executed automated acoustic tests for noise reduction algorithm using MATLAB & C.

Teaching Assistant
Sep 2014 - Dec 2014

Faculty of Engineering, University of Waterloo — Waterloo, Ontario

- Tutored engineering students in C# programming fundamentals.
- Presented course content to students on an individual basis and in classes of up to 100.
- Consistently completed administrative and pedagogical tasks ahead of deadlines.
- Cooperated effectively with teaching team of 18 members to ensure smooth running of the course.

Research Assistant
Jan 2014 - Aug 2014

Center for Intelligent Antenna and Radio Systems, University of Waterloo — Waterloo, Ontario

- Developed and evaluated a ray tracing simulation tool for radio wave propagation using C++ and MATLAB.
- Improved runtime by over 60% using CUDA parallel computing platform on Nvidia GPU hardware.
- Built GUI using MATLAB to visualize simulation results.

Electrical Engineering Intern
May 2013 - Aug 2013

LPH Renewable Energy Ltd. — St Michael, Barbados

- Installed domestic and commercial scale photovoltaic power systems.
- Assisted in project layout calculations and design.
- Streamlined project cost estimation using Microsoft Excel.

EDUCATION

Graduate Certificate, Artificial Intelligence
Jan 2026 - Oct 2026

Seneca Polytechnic — North York, Ontario

- Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing
- GPA: 4.0/4.0
- Honors: President's List

Masters of Engineering, Electrical Engineering
Sep 2017 - Oct 2019

McGill University — Montreal, Quebec

- Relevant coursework: Machine Learning, Signal Processing, Statistical Analysis
- Thesis: Statistical analysis and machine learning algorithms for RF breast cancer screening

Bachelors of Applied Science, Electrical Engineering
Sep 2012 - Jun 2017

University of Waterloo — Waterloo, Ontario

- Co-op program (24 months of industry internships in total).

Secondary School Diploma, Sciences
Sep 2005 - Jun 2012

Harrison College — Bridgetown, Barbados

- CAPE (A-level) Physics, Mathematics & Chemistry

PROJECTS

Personalized Medication Assistant *Sep 2026*

Stack: ***Python, Langchain, ChromaDB, GeminiAPI***
Repo: *github/personalized-medication-assistant*

Smart assistant that uses GeminiAPI and Retrieval Augmented Generation to provide information about user medication regimens. Everyday users can use this intelligent assistant to quickly obtain relevant information about medications stored in their profile.

Text Extraction from Images of Handwritten Prescriptions *Aug 2026*

Stack: ***Python, FastAPI, Docker, GCP, PyTorch, Hugging Face***
Repo: *github/text-from-prescriptions*

Containerized REST API that identifies medication brand names from images of handwritten text.
API can be used to build cloud-deployable web applications to efficiently digitize physical medical records.

Transformer-based Text Classification *Aug 2026*

Stack: ***Python, PyTorch, Hugging Face***
Repo: *github/aig230-project*

NLP pipeline which classifies review sentiments using BERT base model and custom classification head.
Project demonstrates effective application of pre-trained transformer models to custom machine learning tasks.

Gesture Recognition Pipeline *Aug 2026*

Stack: ***Python, Media Pipe, Scikit-Learn***
Repo: *github/aig210-project*

Computer vision pipeline using Media Pipe and a custom classification head to recognize hand gestures in photos.
Project demonstrates effective use of existing machine learning libraries and with custom model development.

Cloud Deployable Computer Vision API *Jun 2026*

Stack: ***Python, FastAPI, PyTorch, Docker, GCP***
Repo: *github/handwritten-digit-api*

Containerized REST API that accepts handwritten digits and returns classifications using a CNN model.
Project demonstrates effective use of data science frameworks, containerization and cloud-deployment tools.

Blog Web Application *Oct 2025*

Stack: ***Bootstrap, EJS, jQuery, Express.js, SQLite***
Demo: collinapjoseph/web-blog
Repo: github/web-blog

Blog application that allows visitors to: create, read, update and delete (CRUD) blog entries.
Project demonstrates effective use of Bootstrap, jQuery, SQL, and understanding of REST design patterns and CRUD workflows.

Tabletop RPG Dice Roller *Oct 2025*

Stack: ***HTML/CSS, jQuery***
Demo: collinapjoseph/ttrpg-dice-roller
Repo: github/ttrpg-dice-roller

Application allows users to make tabletop RPG dice rolls via text input.
Project demonstrates creative application of frontend development knowledge.

ONLINE CERTIFICATIONS

- **Google Prompt Essentials** - Google Career Certificates
  Issued: Aug 2025 | Credential URL: [URL]
- **Deep Learning Specialization** - DeepLearning.AI, Coursera
  Issued: Sep 2019 | Credential URL: [URL]

SKILLS

- Artificial Intelligence
- Machine Learning
- Software Development
- Python
- C#
- C++